

Kumbara by Sembol

• TESTNET

Dollar savings for people paid in lira. Bank transfer in, USDC in a vault, lira back out. A passkey is the only key.

Live (Stellar TESTNET)

kumbara.sembol.xyz

Repository

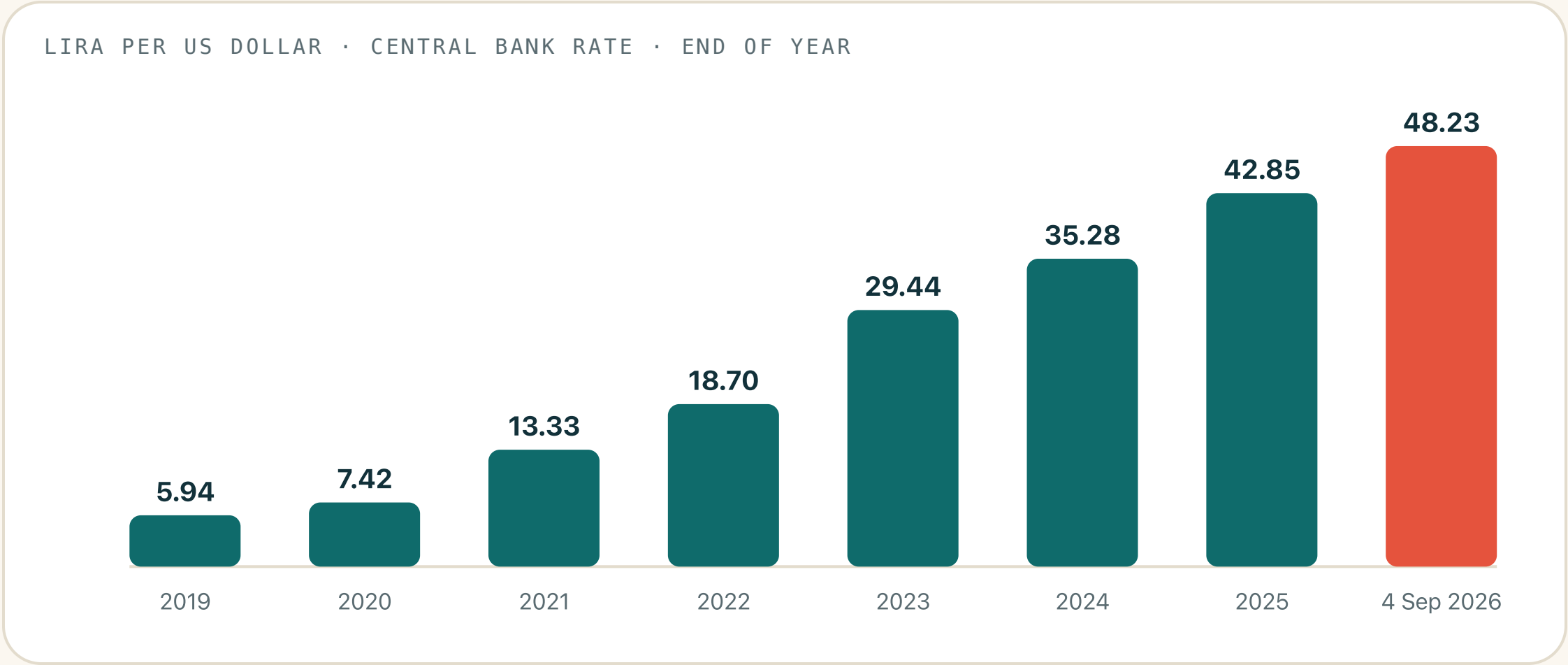
github.com/keyboard01/kumbara

Deck date

7 September 2026

PROBLEM

A lira in the bank buys less every year.



8.1x

the lira price of a dollar since end-2019

31.5%

consumer inflation, August 2026, TÜİK

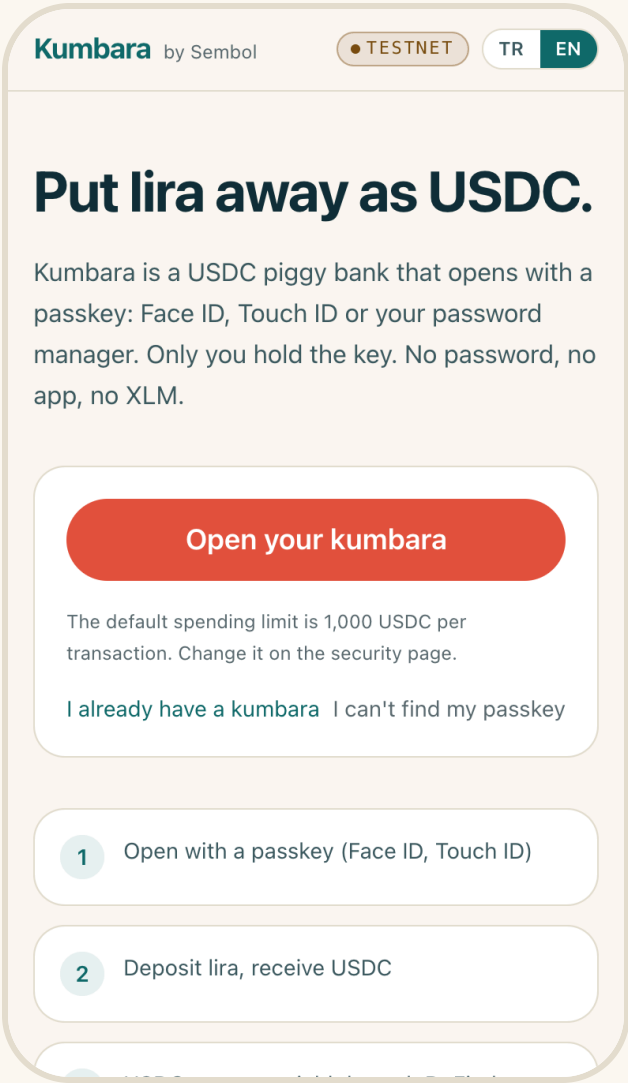
₺100,000 kept as lira since 2019 buys
\$2,073 today. Kept as dollars: **\$16,835**.

The dollar route today: an exchange app, a seed phrase, XLM for fees. Most people stop there.

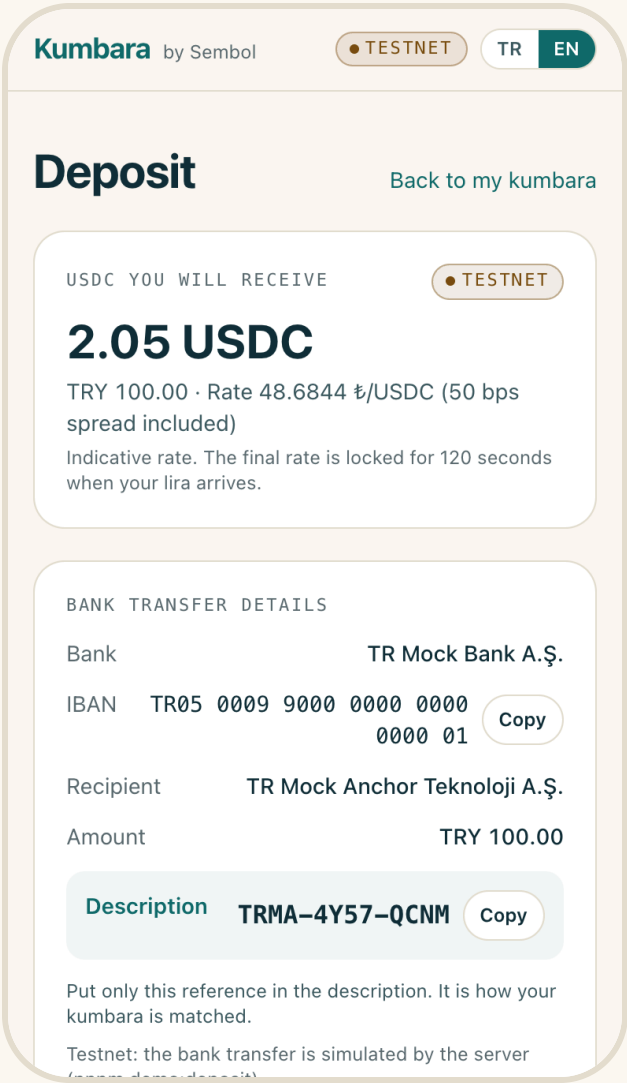
Sources: TCMB indicative USD buying rate, last business day of each year and 4 September 2026 (tcmb.gov.tr/kurlar); TÜİK consumer price index, August 2026, +31.51% year on year (veriportali.tuik.gov.tr/tr/press/58290). Dollar figures are the lira amount divided by those rates.

HOW WE SOLVE IT

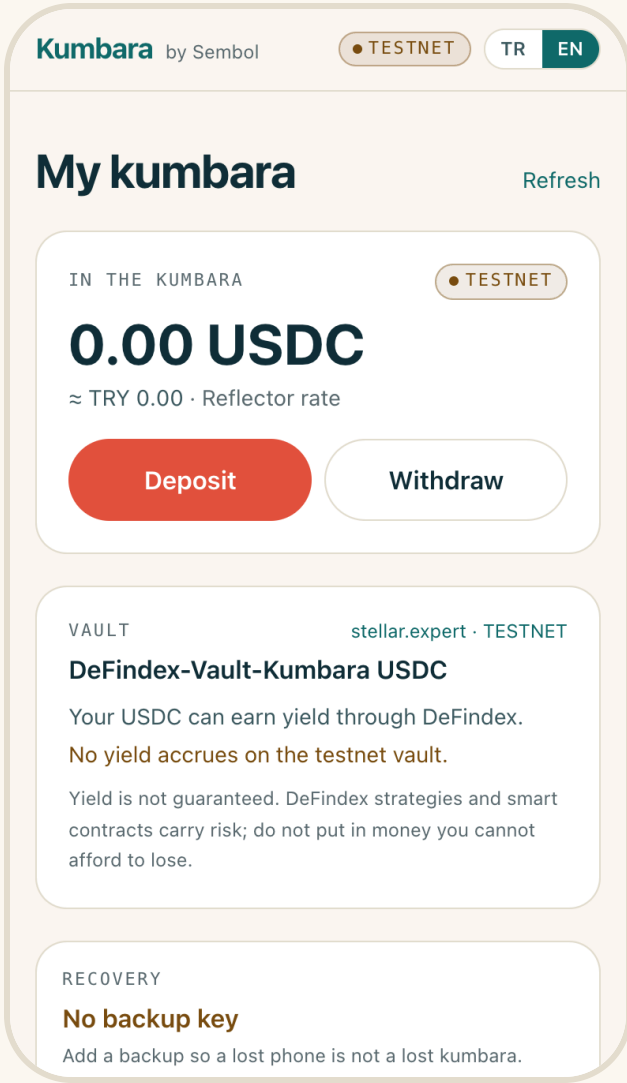
Tap once. Transfer lira. Save in dollars.



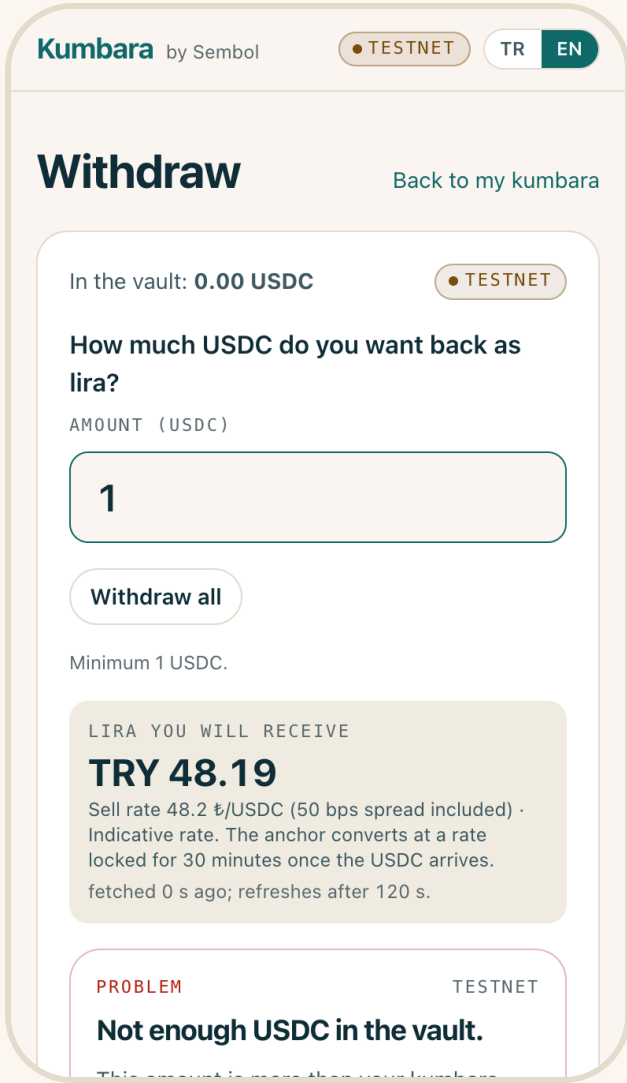
Tap once. A passkey makes the key. No seed phrase, no XLM, no app.



Transfer lira to an IBAN, the way you already pay bills.



Held as USDC in a DeFindex vault. Only you hold the key.



Lira back to your own IBAN, any time.

WHY IT IS BETTER

Self-custody without the crypto homework.

	YOU HOLD THE KEY	NO SEED PHRASE	NO XLM TO BUY	LIRA IN AND OUT BY BANK TRANSFER	NOTHING TO INSTALL
Bank dollar account	✗	✓	✓	✓	✓
Crypto exchange	✗	✓	✓	✓	✗
Stellar wallet app	✓	✗	✗	anchor by anchor	✗
Kumbara	✓	✓	✓	✓	✓

The bank and the exchange hold your dollars. A wallet gives you the key and the homework. Kumbara gives you the key; a passkey signs, a relay pays the fees, a licensed anchor moves the lira.

Five moves, three audited building blocks.

1

Passkey → OpenZeppelin smart account. The relay pays the fees.

2

A 1,000 USDC per-transaction limit, written on-chain.

3

The anchor gives an IBAN. Lira in, USDC out.

4

A bridge account forwards the USDC to your kumbara and closes itself.

5

One approval puts it in the DeFindex vault. Withdrawal runs backwards.

WHAT WE FOUND, WHAT WE BUILT

Anchors pay classic accounts. Smart accounts are contracts. The ramp could not reach the wallet.

Fix: an ownerless bridge account with two pre-signed transactions: forward, then cleanup. Nothing else can ever be signed.

10 / 10

exact amount matches

900

stroops net per deposit

EVIDENCE · STELLAR.EXPERT/EXPLORER/TESTNET/TX/...

Anchor pays the bridge account	778215b3...e3c3a8f1
Pre-signed forward lands in the smart account	f580e030...f22c9eab
Pre-signed cleanup, account merged	6223c61c...7416291d
Vault deposit, invest path	b615e7dc...76b7f981
Vault withdrawal, divest path	466f48d2...07a9f5ca

WHERE ELSE IT WORKS

Same app, next anchor.

MARKET	INFLATION	ANCHOR ON THE SCF INTEGRATION LIST	FIT
Türkiye	34.9% (2025)	sandbox today; licensed TRY anchor = Tranche 1 go/no-go	market one
Argentina	219.9% (2024)	Koywe (sandbox; API in 1–2 weeks)	strong fit
Bolivia	19.5% (2025)	Koywe	strong fit
Nigeria	23.0% (2025)	none by country; global ramps to verify	to verify
Egypt	14.1% (2025)	none by country; global ramps to verify	to verify
Colombia	5.1% (2025)	Koywe	weak fit: low inflation

ADDING A MARKET

- One anchor config: stellar.toml, SEP-6/24 rails, SEP-38 quotes
- One translation file, one currency pair for the rate
- Vault, wallet and bridge accounts do not change
- Our estimate: about 100 hours to wire an anchor sandbox

WHO IT IS FOR

High inflation, a bank-transfer habit, a Stellar anchor. Türkiye first. Argentina and Bolivia next, on the same list.

Inflation: World Bank, consumer prices, annual %, 2025 (Argentina 2024, latest published). Anchors: SCF Integration List, on/off-ramping section, read 6 September 2026.

Built, tested against production, public.

Kumbara by Sembol

TESTNET

TR

EN

Kumbara live

live updated 1:17:47 AM

Public, read-only. Testnet: no real lira moves.

Since event start

All time

1

kumbaras opened

TESTNET

1

deposits completed

TESTNET

0

withdrawals completed

TESTNET

TRY 100.00

TRY in

TESTNET

TRY 0.00

TRY out

TESTNET

26.79

USDC

kumbara.sembol.xyz/stats · live counters, TESTNET badge, seed and test accounts excluded.

6

real-browser round trips in CI, every push and every 6 h against production

25

failure states, each a TR/EN screen with one action

94.100.100.100

Lighthouse mobile on /stats

E2E · passing · production run 7 Sep 2026

GitHub Actions, e2e.yml

Vault deployed with the hodl strategy	dbf0d5b7...6faa8f96
Deposit into the strategy (invest)	b615e7dc...76b7f981
Withdrawal from the strategy (divest)	466f48d2...07a9f5ca

Fallback demo: docs/demo/round-trip.mp4, 2:39, one caption per step.

Non-custodial software in front of a licensed anchor.

- Kumbara never holds lira, USDC or keys.
- The anchor holds the fiat, does KYC, pays your own IBAN.
- No send, no P2P, no merchant flow.
- On-chain per-transaction cap; anchor-side limits.
- Gate: the anchor's written position on a non-custodial front end, Tranche 1.

TÜRKIYE FIRST, BY FIT

31.5% inflation, a transfer habit everyone has, the TRY sandbox this deck runs on.

FALLBACK BUILT IN

Tranche 1 ranks the anchors on the SCF list by market. If Türkiye fails, the pilot launches in the top-ranked market. The app does not change. Market two is the follow-on award.

One builder, already shipping on Stellar.

Ahmed

Founder · builds and pitches

Instaward grantee (Stellar Türkiye). Author of [@sembol/passkey-react](#), the passkey wallet layer Kumbara runs on, and contributor to [stellar/smart-account-kit](#). Designed and built Kumbara end to end: the app, the bridge accounts, the CI round trips and the evidence in this deck.

Site [ahmed.bet](#)

LinkedIn [linkedin.com/in/ahmed-zm](#)

GitHub [github.com/keyboard01](#)

SHIPPED UPSTREAM

stellar/smart-account-kit #4: "Fix wallet creation crash on default Next.js apps", merged 19 August 2026, in release 0.6.2.

PUBLISHED

@sembol/passkey-react on npm: passkey creation and signing, recovery, spending-limit policy UI. Every screen in this deck runs on it.

FUNDED

Instawards grantee. Kumbara is the consumer proof for that work, built for this submission.

Wallet layer and relay hardening first, then testnet-complete and secured, then mainnet in the selected pilot market.

TRANCHE 1 · DAYS 1–60 Wallet layer and relay hardening me 340 h · no hired lines pays 20% at completion (MVP)	- Single-transaction onboarding: spending-limit policy on the default rule at creation; the library's limit UI updated @sembol/passkey-react 0.4 published with projectId: relay project binding, config, docs, migration notes - Relay mainnet hardening of the fee-sponsoring path: per-project budgets, allowlists, refusal alerts, sponsor monitoring - Anchor shortlist and go/no-go: on/off-ramps ranked by market, written positions from the top two, pilot anchor chosen, runner-up named - Audit Bank application submitted with scope: policy install, bridge accounts, server path Shortlist input, unranked until this tranche (SCF Integration List, on/off-ramping): Koywe · alfredpay · Abroad · BlindPay · Mercuryo · Bridge. No TRY anchor is listed today; Türkiye stays market one only if a licensed TRY anchor passes the same criteria, otherwise the pilot launches in the top-ranked market and the app does not change. The runner-up is a shortlist output, not an integration in this award.
TRANCHE 2 · DAYS 61–120 Testnet-complete and secured me 340 h + hired 80 h pays 30% at completion (testnet)	- Selected anchor wired in via its sandbox or API in place of the mock anchor (SEP-6/24 and SEP-38 where offered) - Threat model in SDF's STRIDE template format: assets, threat table, mitigations - Monitoring plan (builders template) with core alerts live for vault, policy, sponsor and bridge accounts; dashboards hired, optional - Audit Bank audit of the policy install, bridge accounts and server path; findings remediated with regression tests; QA hired - Mainnet path behind a capped, presenter-only flag with real USDC; hosted and self-host deployment docs with scripted setup
TRANCHE 3 · DAYS 121–180 Mainnet launch and public pilot me 340 h + hired 80 h pays 40% at completion (mainnet)	- Mainnet launch in the pilot market by week 3: contracts registered, sponsor and relay budgets live, production rails - Public pilot cohort of 50–100 users; validation-focused user testing run by me; session ops and UX polish hired (1/3 exception) - Monitoring stack live on mainnet with an on-call runbook: rotation, escalation, incident templates - Metric window: 30-day NAV measurement, public dashboard, on-chain footprint registered - Pilot support process in operation: support channel, triage rules, response targets, fixes, weekly pilot reports

Boundary with the Sembol Instaward. Instaward Month 3 delivers the testnet relay. This award funds only what lies beyond it: the mainnet hardening of that fee-sponsoring path (per-project budgets, allowlists, refusal alerts, sponsor monitoring), the 0.4 library release and policy-based single-transaction onboarding. No hour appears in both.

BUDGET AND METRIC · \$107,300 · INTEGRATION TRACK LARGE BAND (\$100K–150K), AT ITS LOW END · EVERY DOLLAR IS HOURS

One full-time engineer plus hired capacity: 1,180 hours over six months, \$107,300.

TRANCHE	DAYS	ME H	HIRED H	ME USD	HIRED USD	SUBTOTAL
1 · Wallet layer and relay hardening	1–60	340	0	\$32,300	\$0	\$32,300
2 · Testnet-complete and secured	61–120	340	80	\$32,300	\$5,200	\$37,500
3 · Mainnet launch and public pilot	121–180	340	80	\$32,300	\$5,200	\$37,500
Total · six months	180	1,020	160	\$96,900	\$10,400	\$107,300

Paid in XLM per the handbook split: 10% on acceptance \$10,730 · 20% after Tranche 1 \$21,460 · 30% after Tranche 2 \$32,190 · 40% after Tranche 3 \$42,920

HIRED CAPACITY · 160 H AT \$65/H · PARALLEL WORK ONLY; NO TRANCHE GATE DEPENDS ON A HIRE

Monitoring dashboards and alert routing, optional (Tranche 2)	40 h	\$2,600
QA against the audit findings: independent re-test, regression suite (Tranche 2)	40 h	\$2,600
User-testing ops: structured sessions, interviews, write-ups (Tranche 3)	40 h	\$2,600
UX polish for the pilot: copy, edge states, accessibility fixes (Tranche 3)	40 h	\$2,600
No hire, no bill: each tranche closes on my own lines. Dashboards are optional by design.		

FINAL-TRANCHE METRIC COMMITMENT

Metric type	DeFindex vault NAV: USDC in Kumbara's mainnet vault position, from local-currency deposits in the pilot market
Threshold	average daily NAV of 2,500 USDC = 100 pilot users × 50 USDC average balance × 50% still holding through the window. The cohort is planned at 50–100 users; the threshold takes the top of that range
Window	the 30 days after the pilot launch, averaged over the window, not a snapshot
Footprint, registered at award time	vault, strategy and policy contracts; sponsor account (bridge accounts sponsored per CAP-33); app-created smart accounts

The panel ratifies or adjusts the threshold during panel review. Seed and E2E accounts are tagged out of every number; wash or gamed activity is excluded.

BUDGET HYGIENE

Development, testing, monitoring, deployment, validation. Nothing self-funded; every line is hours × rate. No legal fees, no entity setup, no audit fee (Audit Bank), no marketing beyond the user-testing exception (about 8% of the total). Nothing depends on a third party signing. Nothing here is paid by the Instaward.

Paid in XLM in tranches of 10% on acceptance, 20% at MVP, 30% at testnet and 40% at mainnet; "Project timeline should not exceed 6 months"; each tranche completion form within 90 calendar days of the previous payment: SCF Handbook, Build Award, "Budget & Deliverable Guidelines" (stellar.gitbook.io/scf-handbook/scf-awards/build-award/budget-and-deliverable-guidelines). Rates: \$95/h for a senior full-stack engineer with Soroban and passkey work shipped upstream (stellar/smart-account-kit PR #4; @sembol/passkey-react on npm), full-time at 40 h/week, 340 h per 60-day tranche; hired capacity at \$65/h for parallel work only.

20 lines, 1,180 hours, \$107,300. Each line names how reviewers verify it.

DELIVERABLE	HOW REVIEWERS VERIFY	WHO · HOURS	USD
TRANCHE 1 · WALLET LAYER AND RELAY HARDENING · DAYS 1–60		ME 340 H	\$32,300
Single-transaction onboarding: the spending-limit policy installed on the default rule at account creation; the library's limit UI updated	one tx hash per new kumbara on testnet; onboarding timing on /stats	me · 90 h	\$8,550
@sembol/passkey-react 0.4 published with projectId: relay project binding, config, docs, migration notes; Kumbara moved onto it	npm 0.4.x; changelog; Kumbara lockfile	me · 70 h	\$6,650
Relay mainnet hardening of the fee-sponsoring path: per-project budgets, allowlists, refusal alerts, sponsor monitoring	config in repo; a refused call's log and tx; an alert fired; sponsor monitor	me · 100 h	\$9,500
Anchor shortlist and go/no-go: Integration List on/off-ramps ranked by market; written positions from the top two; pilot anchor and market selected; runner-up named as the fallback	ranked shortlist, decision memo, written positions (repo)	me · 60 h	\$5,700
Audit Bank application with scope: policy install, bridge accounts, server path	submission confirmation; scope document	me · 20 h	\$1,900
TRANCHE 2 · TESTNET-COMPLETE AND SECURED · DAYS 61–120		ME 340 H + HIRED 80 H	\$37,500
Selected anchor wired in via its sandbox or API in place of the mock anchor (SEP-6/24, SEP-38 where offered; adapter otherwise)	sandbox round trip green in CI	me · 100 h	\$9,500
Threat model in SDF's STRIDE template format (assets, threat table, mitigations)	docs/security/threat-model.md	me · 50 h	\$4,750
Monitoring plan in the builders template; core alerts implemented for the vault, policy, sponsor and bridge accounts	monitoring-plan.md; a live alert on testnet	me · 50 h	\$4,750
↳ Monitoring dashboards and alert routing, optional: boards per contract and account, paging rules	dashboard URL; a routed alert	hired · 40 h	\$2,600
Audit Bank audit of the policy install, bridge accounts and server path: scoping, support, findings remediated with regression tests (fee: Audit Bank)	audit report; fix commits; auditor re-check	me · 80 h	\$7,600
↳ QA against the audit findings: independent re-test, regression suite extended	QA report; CI suite	hired · 40 h	\$2,600
Mainnet path behind a capped, presenter-only flag: real USDC, enforced budgets	mainnet tx hashes from the capped path	me · 30 h	\$2,850
Hosted (Vercel + Turso) and self-host (Fly + SQLite) deployment docs with scripted setup; CI against the anchor sandbox	docs URLs; a fresh deploy from the docs; CI run	me · 30 h	\$2,850
TRANCHE 3 · MAINNET LAUNCH AND PUBLIC PILOT · DAYS 121–180		ME 340 H + HIRED 80 H	\$37,500
Mainnet launch in the selected pilot market by week 3: contracts registered, sponsor account and relay budgets live, production rails	mainnet contract IDs on stellar.expert	me · 90 h	\$8,550
Public pilot cohort of 50–100 users; validation-focused user testing, run by me (within the 1/3 exception)	/stats on mainnet; tx hashes; cohort report	me · 60 h	\$5,700
↳ User-testing ops: structured sessions, interviews, write-ups	session log; cohort report	hired · 40 h	\$2,600
↳ UX polish for the pilot: copy, edge states, accessibility fixes from the sessions	fix commits; Lighthouse accessibility score	hired · 40 h	\$2,600
Monitoring stack live on mainnet with an on-call runbook: rotation, escalation, incident templates	runbook in repo; a mainnet alert and its ticket	me · 50 h	\$4,750
Metric window: 30-day NAV measurement, public dashboard, on-chain footprint registered	NAV dashboard URL; registered IDs	me · 40 h	\$3,800
Pilot support process in operation: support channel, triage rules, response targets, fixes, weekly pilot reports	issue log; weekly reports in the repo	me · 100 h	\$9,500
Total · Integration Track Large band (\$100k–150k), at its low end · me 1,020 h (\$96,900) + hired 160 h (\$10,400)		1,180 h	\$107,300